AI-based energy management systems

"AI-based energy demand forecasting and Smart microgrids and energy management" *Generated by LeapSpace, Sat Oct 10 2026*

AI-based energy demand forecasting is a strong enabler of smart microgrids and energy management because it improves prediction accuracy, which in turn supports better scheduling, demand response, storage use, and real-time control. The evidence also shows that combining forecasting with optimization or control methods such as MPC, reinforcement learning, and hybrid metaheuristics is a recurring pattern for improving cost, reliability, and renewable integration in microgrids. [1] [2] [3] [4] [5] [6] [7] [8] [9]

What the evidence says

Theme	What it shows	Why it matters
Demand forecasting	AI/ML models such as RF, XGBoost, LSTM, CNN-LSTM, ANN, and hybrid deep learning approaches consistently improve short-term or day-ahead demand forecasting accuracy in smart-grid and microgrid settings. [5] [6] [7] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22]	Better forecasts reduce uncertainty in load balancing and improve dispatch decisions. [1] [10] [23]
Microgrid energy management	Forecasts are repeatedly used to optimize energy scheduling, battery operation, renewable utilization, and peak shaving in smart microgrids. [3] [4] [8] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [37]	This connects prediction directly to lower operating cost, better reliability, and higher renewable penetration. [4] [24] [27] [28]
Demand response and pricing	Demand response, time-of-use pricing, and incentive schemes are common levers for shaping demand and reducing peaks in microgrids. [7] [24] [34] [38] [39] [40] [41] [42] [43] [44] [45] [46]	Forecasting helps decide when and how to shift demand, which improves load flexibility. [7] [24] [34]
Optimization and control	MPC, reinforcement learning, stochastic/robust optimization, and metaheuristics are widely used to turn forecasts into operational decisions. [3] [12] [24] [29] [30] [36] [47] [48] [49] [50] [51] [52]	Forecasting alone is not enough; control logic is what converts prediction into actionable energy management. [4] [36] [49] [50]

Theme	What it shows	Why it matters
Real-world performance gains	Several studies report measurable improvements such as lower MAE/RMSE, reduced costs, lower emissions, improved stability, and better storage utilization. [6] [8] [13] [14] [17] [18] [24] [26] [28] [31] [39] [53] [54]	This suggests AI-based forecasting is practically useful, not just theoretically appealing. [2] [4] [9]

Bottom line

The clearest pattern is that **AI-based demand forecasting is most valuable when embedded in a broader microgrid management stack**: forecast models estimate demand and renewable output, then optimization, demand response, storage scheduling, and control algorithms use those predictions to cut cost, smooth peaks, and improve resilience. [1] [3] [4] [6] [8] [24] [30] In other words, the literature treats forecasting as the front end of intelligent energy management rather than a standalone task. [2] [23] [55]

Confidence level: High — multiple independent sources converge on the same architecture and benefits across forecasting, demand response, optimization, and microgrid operation.

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AI-driven energy management in microgrids

"AI-based energy demand forecasting and Smart microgrids and energy management integreted with Smart EV charging and load management and and load management" *Generated by LeapSpace, Sat Oct 10 2026*

AI-based **energy demand forecasting** is a strong fit for **smart microgrids** when it is paired with **EV charging control, demand-side management, and load management**, because the forecasting layer improves scheduling while smart charging and DSM convert those predictions into lower peaks, lower costs, and better grid stability [1] [3] [29] [56] [57] [58] [59] [60] [61] [62].

How the pieces fit together

Component	What the evidence shows	Why it matters
AI demand forecasting	AI/ML models improve load and demand prediction in smart grids and microgrids, including LSTM, XGBoost, DNN, CNN-GRU, and hybrid models [1] [11] [16] [59] [63] [64] [65] [66] [67] [68] [69] [70].	Better forecasts support more accurate scheduling, reserve planning, and real-time control [1] [59] [71].
Microgrid energy management	Forecasts are used with MPC, reinforcement learning, and other optimization methods to reduce operating cost and improve microgrid reliability and renewable utilization [1] [3] [29] [57] [60] [72] [73] [74] [75].	Forecasting becomes operational value only when linked to control and dispatch [1] [72] [75].
Smart EV charging	Smart charging, V1G, and V2G are repeatedly described as flexible-load tools that shift charging to better grid conditions, reduce peak demand, and enable bidirectional energy exchange [60] [61] [62] [76] [77] [78] [79] [80] [81] [82] [83] [84] [85] [86] [87] [88] [89] [90] [91] [92] [93] [94] [95].	EVs can be treated as controllable demand or distributed storage, which helps peak shaving and load balancing [60] [61] [79] [80] [86].
Load management / DSM	Demand response, load shifting, and load curtailment are central mechanisms for reducing peak hours and balancing supply-demand mismatch in microgrids [1] [3] [76] [78] [85] [87] [91] [92] [96] [97] [98].	DSM is the bridge between forecasting and action because it turns predicted demand into controllable behavior [1] [98].

What the literature most clearly supports

- **Forecasting + EV charging control** is the most common integration pattern: predicted demand and price signals are used to schedule charging/discharging, reduce peak stress, and improve load factor [61] [62] [76] [77] [80] [81] [87].
- **V2G and smart charging** consistently improve flexibility by letting EVs act as mobile storage or adjustable loads, which helps with peak shaving and renewable absorption [60] [79] [80] [85] [86] [88] [91] [99] [100].
- **AI-based EMS architectures** are increasingly hierarchical: forecasting at the top, MPC or RL in the middle, and device-level scheduling at the bottom [29] [57] [72] [75]. This is an inference from the repeated structure across studies rather than a single explicit standard [29] [57] [72] [75].
- **Measured benefits** include lower operating cost, lower peak demand, improved voltage profiles, reduced losses, better renewable utilization, and more stable microgrid operation [1] [57] [58] [61] [62] [77].

Bottom line

The strongest supported view is that **AI demand forecasting is the prediction layer, and smart EV charging plus DSM are the execution layer** of modern microgrid energy management [1] [57] [60] [62] [76]. In practice, the best-performing systems do not treat forecasting, EV charging, and load management as separate problems; they combine them so forecasts guide charging schedules, load shifting, storage dispatch, and real-time control to cut cost and peak stress while improving resilience [1] [3] [29] [61] [72] [75].

Confidence level: High — multiple independent sources consistently converge on the same integrated architecture and report similar operational benefits across forecasting, smart charging, and load management [1] [57] [60] [62] [76] [77].

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Research gaps in integrated microgrid management

"What are the research gaps" *Generated by LeapSpace, Sat Oct 10 2026*

The main research gaps are **real-world validation, standardization, scalability, interoperability, and human-centered deployment**—with additional gaps around **EV integration, uncertainty handling, explainability, privacy/security, and long-term economic/policy assessment** in AI-enabled microgrids and energy management [1] [101] [102] [103] [104] [105] [106] [107] [108] [109].

Research gaps by theme

Gap area	What is still missing	Why it matters
Real-world deployment	Many studies still rely on simulation, and several call for field testing, documented case studies, and practical implementation under real operating conditions [101] [103] [110] [111].	Without deployment evidence, feasibility, reliability, and operational value remain uncertain [101] [103] [104].
Scalability and multi-microgrid coordination	Validation on larger grids, smart cities, national networks, and multi-microgrid coordination remains limited [102] [103] [108] [112].	Many methods may work at small scale but not under broader network complexity [102] [108] [112].
Standardized benchmarking	There is a repeated need for common evaluation metrics, performance benchmarks, and standardized validation platforms [101] [105] [108] [113] [114].	Lack of standardization makes it hard to compare methods fairly or replicate results [101] [113] [114].
EV and flexible-load integration	Multiple sources identify EV charging, demand response, and flexible-load coordination as underdeveloped areas for microgrids [105] [111] [115] [116] [117] [118] [119].	EVs can strongly reshape peak demand, so ignoring them leaves an important operational gap [115] [116] [117].
Uncertainty and real-world operating constraints	Real-world conditions such as demand uncertainty, tariffs, export limits, communication latency, sensor error, and data reliability are not fully addressed [103] [104] [107] [111] [113].	These constraints are central to whether AI controllers work outside controlled settings [103] [104].
Explainability and trust	Many AI forecasting and control models still lack transparency, interpretability, and actionable explanations [107] [120] [121] [122] [123] [124] [125].	Black-box performance is not enough for critical energy decisions that require trust and accountability [122] [125].

Gap area	What is still missing	Why it matters
Privacy, cybersecurity, and governance	Privacy-preserving methods, cybersecurity, regulatory frameworks, and governance mechanisms are still unevenly developed [108] [109] [125] [126] [127].	Energy AI systems need secure and policy-compatible deployment, not just technical accuracy [109] [127].
Economic viability and user adoption	Several studies call for stronger evidence on costs, return on investment, business models, and user accessibility [102] [111] [123] [128].	Adoption depends on affordability and usability, not only technical performance [102] [128].

Bottom line

The literature is already strong on **methods** and **performance gains**, but weaker on **how these systems behave in messy, real, interconnected deployments**—especially when forecasting is combined with EV charging, demand response, and microgrid control [1] [103] [104] [108] [119]. In practical terms, the next research wave needs to move from “does the model work?” to “does the integrated system remain reliable, explainable, secure, and economically viable at scale?” [101] [102] [107] [114] [125].

Confidence level: High — multiple independent reviews and conclusion sections converge on the same gap pattern across microgrids, forecasting, EV integration, and AI-enabled energy management [101] [105] [107] [108] [122] [129].

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Explainable AI in energy forecasting

"How do explainable AI methods support trust in energy demand forecasting?" *Generated by LeapSpace, Sat Oct 10 2026*

Explainable AI supports trust in **energy demand forecasting** by making model outputs **transparent, interpretable, and auditable**, so users can see *why* a forecast was produced instead of relying on a black box. In the retrieved energy literature, this is mainly achieved through **feature attribution methods** like SHAP and LIME, **intrinsically interpretable models** like fuzzy logic systems, and **user-centered explanations** that connect predictions to actionable energy variables and decisions [56] [122] [130] [131] [132] [133] [134] [135].

How XAI builds trust in forecasting

Trust mechanism	How it helps	Evidence
Transparency of predictions	XAI reveals which inputs most influenced a demand forecast, reducing the "black-box" problem and making the model easier to inspect [56] [130] [131] [133].	SHAP, LIME, DeepLIFT, attention-based visualization, and fuzzy-rule explanations are all used to expose feature importance and model reasoning [56] [130] [131] [132] [135].
User understanding	Explanations help operators, managers, and end-users understand how variables like weather, time, consumption patterns, and equipment states drive forecasts [56] [134] [136] [137] [138].	This makes forecasts easier to validate against domain knowledge and practical experience [137] [139] [140].
Accountability and compliance	XAI supports traceability, transparency, and regulatory alignment, which are important in critical energy systems [134] [135] [141] [142] [143].	The literature repeatedly links explainability with trustworthy deployment in smart grids and forecasting [56] [135] [141].
Actionable decision support	Explanations are most useful when they are tied to operational choices such as demand response, load shifting, and energy allocation rather than just feature plots [134] [137] [144].	This makes forecasts more actionable for grid operators and planners [56] [134] [136].
Robustness and validation	Some XAI methods are used for feature selection or rule pruning, improving interpretability without necessarily sacrificing accuracy [132] [133] [145].	In one energy forecasting example, SHAP-guided feature reduction with an interval type-2 fuzzy system improved RMSE while keeping the model interpretable [132].

Which methods matter most

- **SHAP** is the most common trust-building method in the evidence because it provides both **global and local** explanations and is widely used for energy forecasting interpretability [122] [130] [131] [132] [133] [135] [145] [146].
- **LIME** is valued for **local, case-by-case explanations**, especially when users want to know why a specific forecast was made [131] [135] [147] [148] [149].
- **Counterfactual explanations** help by showing how small input changes would alter the forecast, which aligns with human reasoning and supports trust [135] [149].
- **Fuzzy logic and rule-based XAI** increase trust by embedding interpretability into the model itself, rather than only explaining it afterward [56] [132] [150].
- **Clustering-assisted SHAP** can reduce computational cost while keeping explanations stable, which matters for practical deployment [145].

Bottom line

XAI supports trust in energy demand forecasting by turning forecasts from **opaque predictions** into **inspectable decisions**: it shows which factors mattered, helps users check whether the result matches domain expectations, and makes the system more defensible for operational and regulatory use [56] [130] [134] [135] [137]. The strongest pattern in the evidence is that trust increases most when explainability is **user-centered and operationally meaningful**, not just a post-hoc visualization layer [131] [134] [144] [146].

Confidence level: **High** — multiple independent sources consistently describe XAI as improving transparency, interpretability, accountability, and user trust in energy forecasting, with SHAP, LIME, and fuzzy-logic explanations recurring across the evidence [56] [122] [130] [131] [135].

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